2-W1　有限带电棒的一般场点

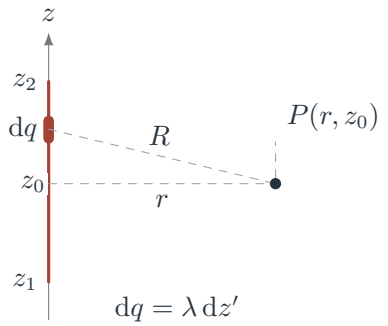


图 2-18　题图　有限棒的一般场点与源微元

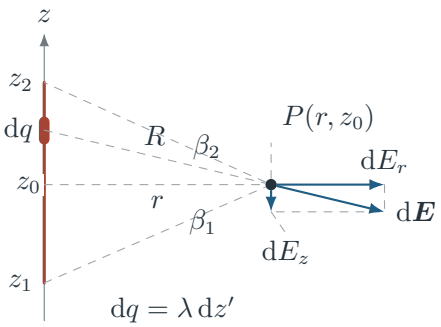


图 2-19　解答分析图　图示 $\lambda > 0$；源到场点位移、端点角与分量

2-W2　非均匀带电棒的电场

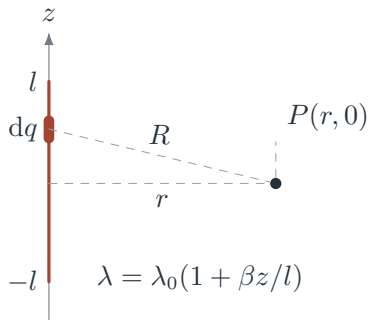
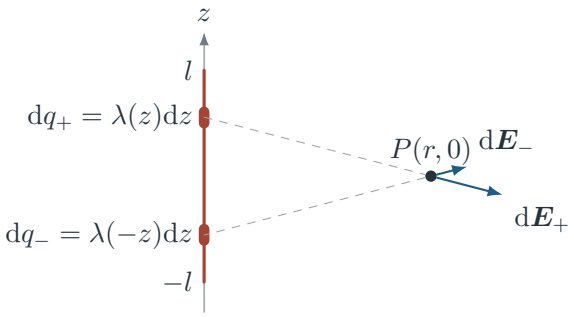


图 2-20　题图　密度随 z 变化的对称棒



图示 $\beta > 0, z > 0$：上侧微元电量更大

E_r 保留密度偶部，E_z 保留密度奇部

图 2-21　解答分析图　上、下对应微元不等，轴向不能全抵消

2-W3　均匀圆弧与带角度变化的圆环

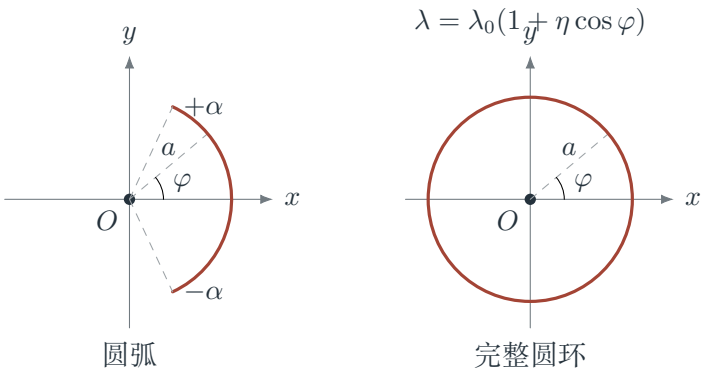
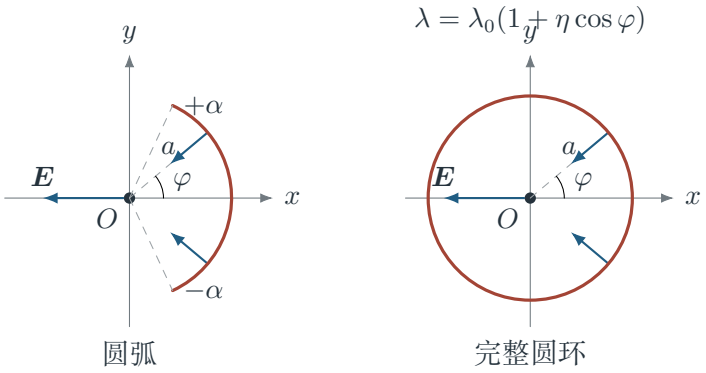


图 2-22　题图　圆弧与完整非均匀圆环；两分问的密度与对称性分别处理



箭头取 $\lambda>0$、$\lambda_0>0$、$\eta>0$；上下微元配对

图 2-23　解答分析图　圆心微元场与角度投影；完整环另按密度积分

2-W4　圆环轴线：最大场强与约束下的小振动

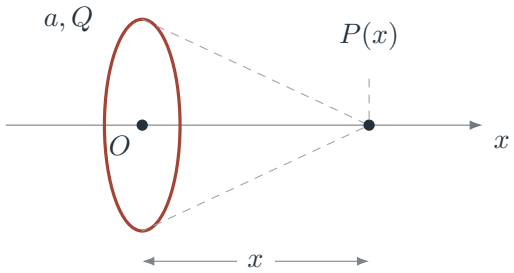
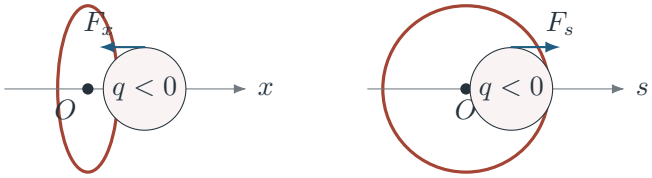


图 2-24　题图　均匀圆环与轴线场点



$x>0$：轴向恢复　　$s>0$：平面内失稳

$F_x \simeq -kQ|q|x/a^3,$　　$F_s \simeq +kQ|q|s/(2a^3)$

图 2-25　解答分析图　近环心的负粒子：轴向恢复、环平面内失稳；撤去导轨后不能三维稳定。

2-W5　从圆环积分得到圆盘

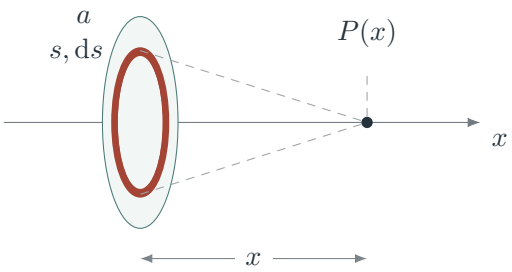


图 2-26　题图　圆盘划分成同心细环

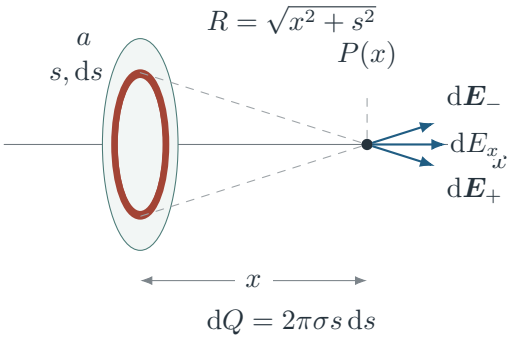


图 2-27　解答分析图　图示 $\sigma>0$；取环微元后只积分一个径向变量

2-W6　带孔圆盘与叠加法

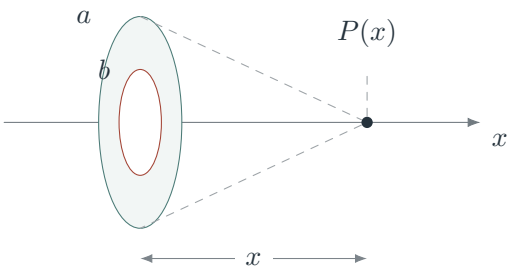
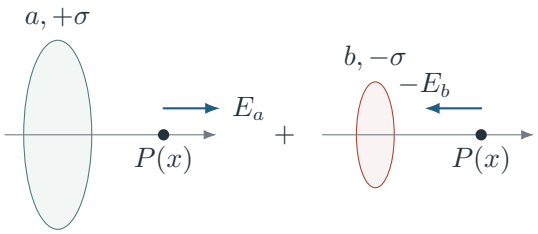


图 2-28　题图　有孔圆盘的轴线几何



$$E_{\text{环面}} = E_a - E_b \quad (E_b \text{ 指正密度内盘的场})$$

$$E_x = \frac{\sigma x}{2\varepsilon_0}\left[(x^2+b^2)^{-1/2} - (x^2+a^2)^{-1/2}\right]$$

图 2-29　解答分析图　外盘减内盘；小位移须满足 $|x| \ll b$

2-W7　电偶极子的一般远场与近似精度

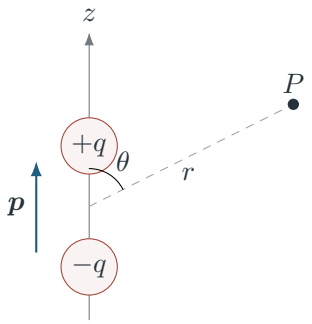


图 2-30　题图　电偶极子与一般方向场点

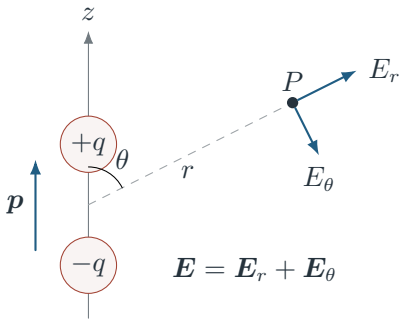


图 2-31　解答分析图　径向与极角方向分量的合成

2-W8　点电荷通过开曲面的通量

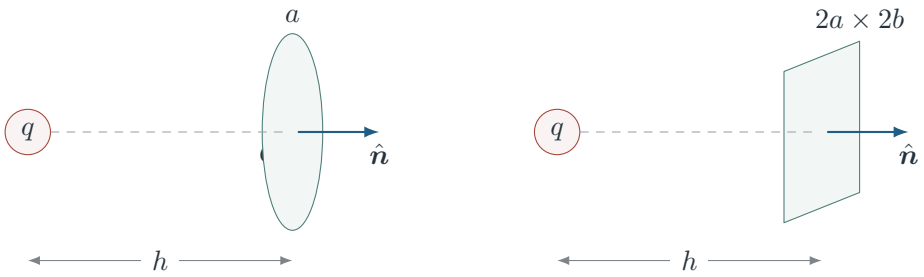


图 2-32　题图　圆盘与矩形开曲面；法向均取背离点源

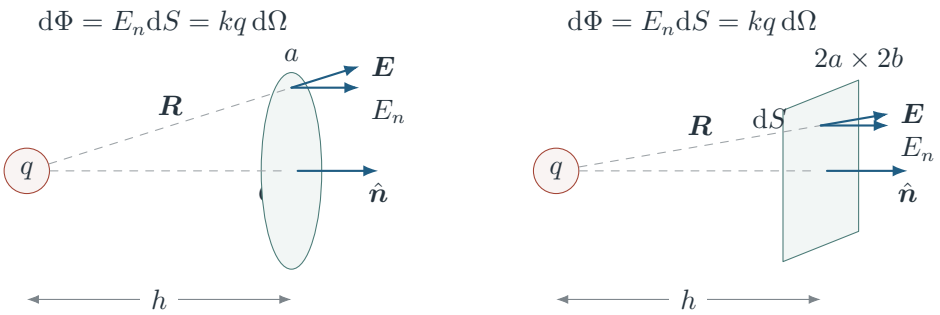
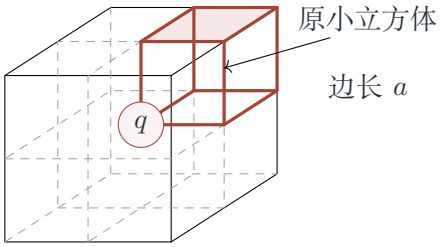


图 2-33　解答分析图　圆盘与矩形的法向投影采用同一几何核；分别作面积积分。

2-W9　立方体的通量与边界奇点



图 2-34　题图　点电荷位于立方体中心或顶点的两种题设



大立方体边长 $2a$；每远端小面占大面 $1/4$

图 2-35　解答分析图　中心情形等分六面；顶点情形用八体拼接

2-W10　由已知球对称场反演电荷分布

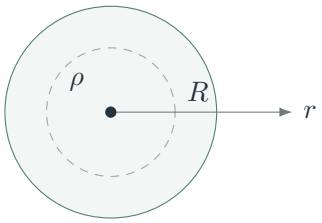
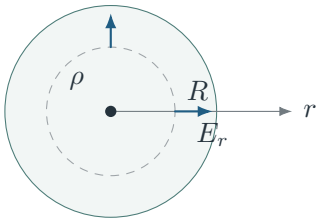


图 2-36　题图　球对称场与同心高斯面



$$Q_{内} = 4\pi\varepsilon_0 r^2 E_r$$

$$\mathrm{d}Q = 4\pi r^2 \rho \,\mathrm{d}r$$

图 2-37　解答分析图　包围电量的变化反映体密度

2-W11　油滴实验：从两次终速反演电荷

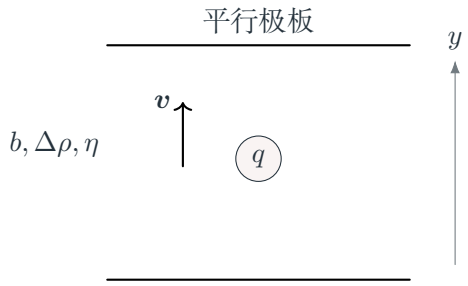
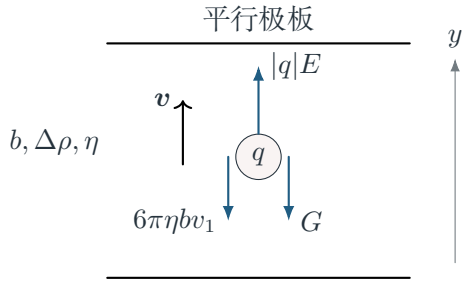


图 2-38　题图　加场后向上运动的油滴



向上终速：重力与阻力都向下

图 2-39　解答分析图　向上终速时的三个受力方向

2-W12　总电荷和偶极矩都为零：首个非零远场

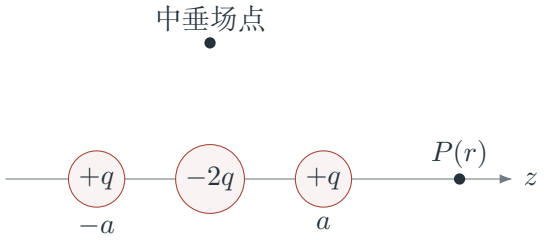


图 2-40　题图　总电量与偶极矩都为零的三电荷

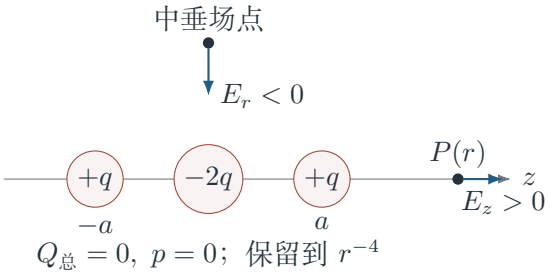


图 2-41　解答分析图　不同观测方向的四极远场

2-T1　非均匀电荷密度的归一化

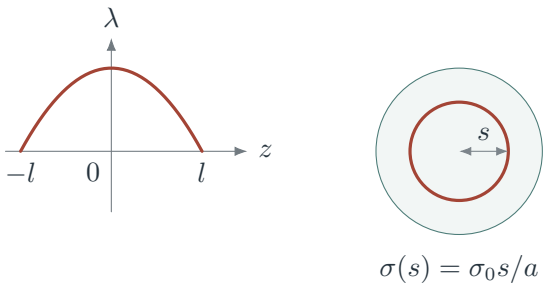


图 2-42　题图　非均匀细棒与径向非均匀圆盘

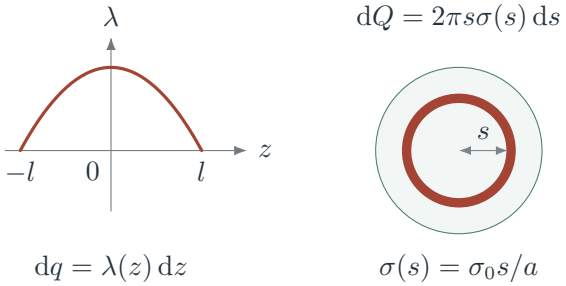


图 2-60　解答分析图　线微元与细环面积微元

2-T2　对称点电荷的电场与极值

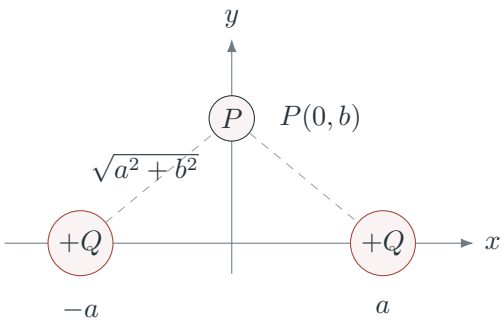


图 2-43　题图　两个正源的中垂线场点

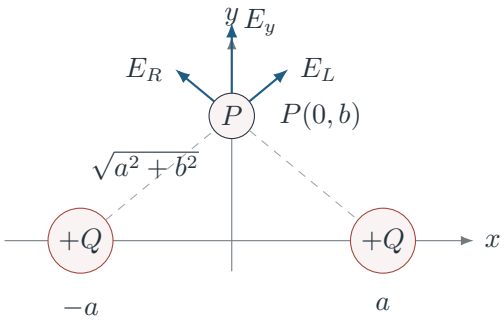


图 2-61　解答分析图　横向抵消，纵向投影相加

2-T3　半无限长带电直线

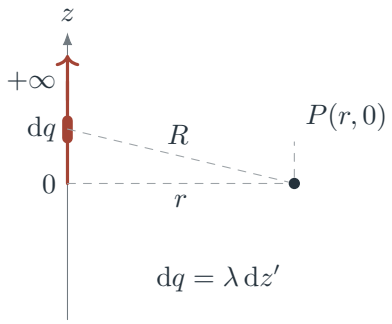


图 2-44　题图　从原点向上延伸的半无限细线

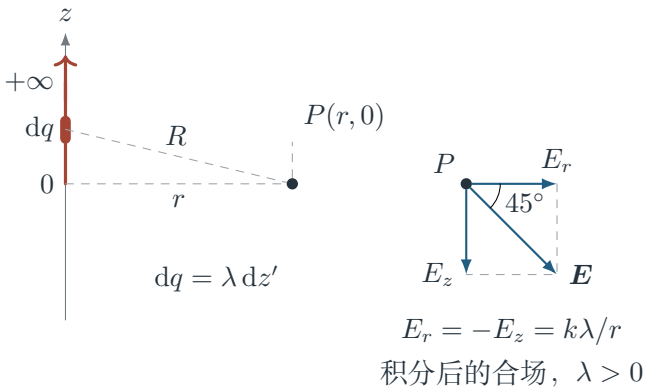


图 2-62　解答分析图　半无限棒的场积分后作等分量合成

2-T4　通量是带符号的投影

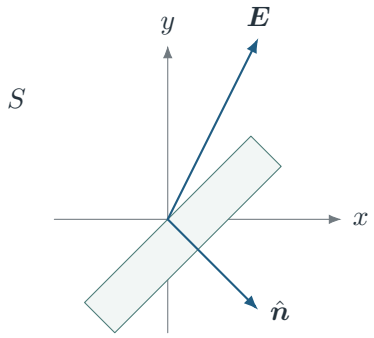


图 2-45　题图　匀强场、面积及指定法矢的截面示意

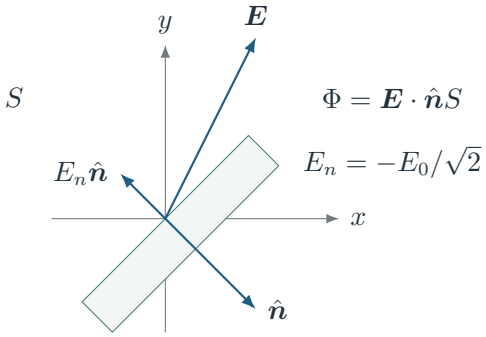


图 2-63　解答分析图　投影可为负，通量随法向反转

2-T5　上半圆的中心场

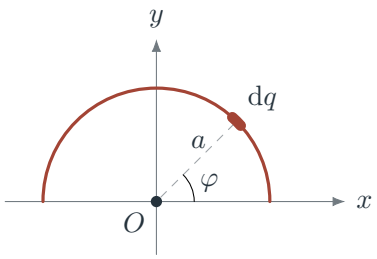


图 2-46　题图　上半圆与中心场点

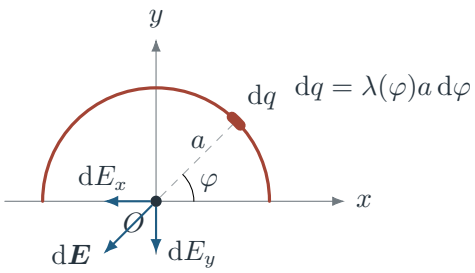


图 2-64　解答分析图　微元场向圆心，密度变化时分量分别积分

2-T6　非均匀棒的远场方向

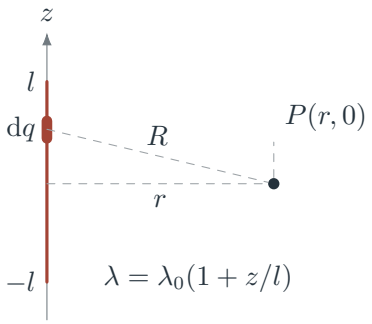
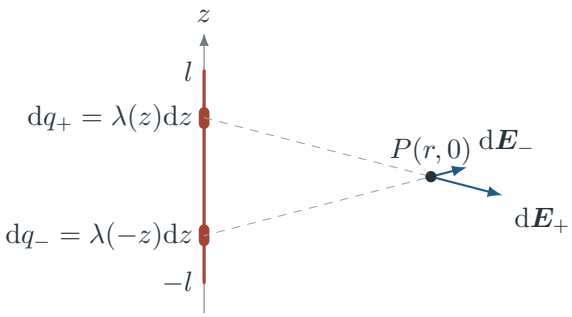


图 2-47　题图　非均匀棒的中垂面远场点；取 $\beta = 1$



图示 $\beta = 1, z > 0$：上侧微元电量更大
E_r 保留密度偶部，E_z 保留密度奇部

图 2-65　解答分析图　径向总电荷项与轴向偶极项

2-T7　轴线上零场不等于空间处处零场

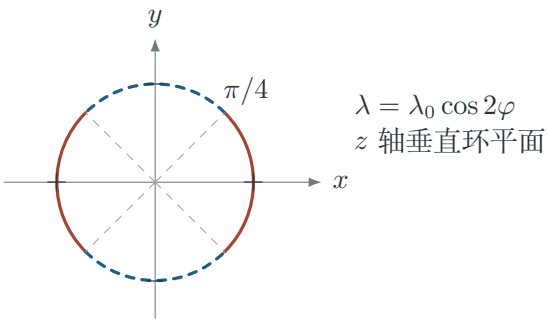


图 2-48　题图　环上密度的正负区域

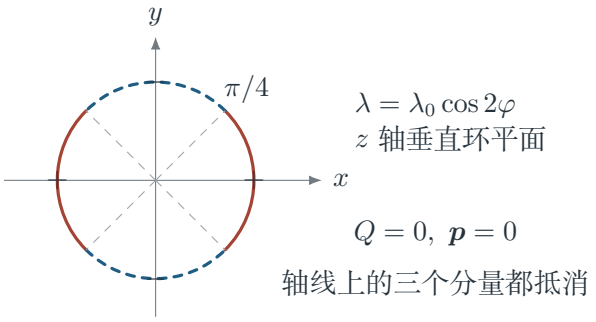


图 2-66　解答分析图　轴线角积分抵消，空间其他点不一定零场

2-T8　圆盘与点电荷近似

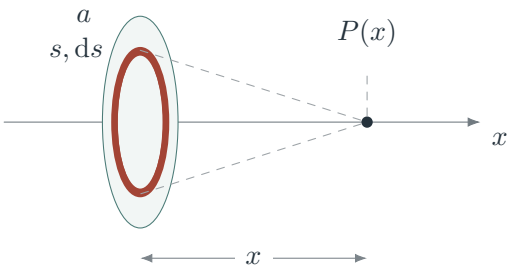


图 2-49　题图　圆盘轴线上的近、远场观测

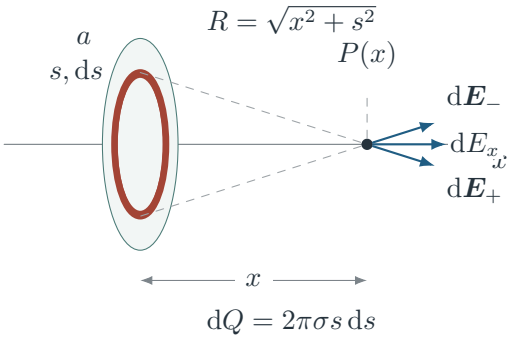


图 2-67　解答分析图　细环积分与点电荷近似的首阶修正

2-T9　圆环形带电面的最大场

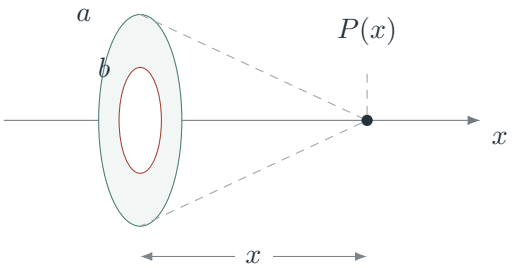
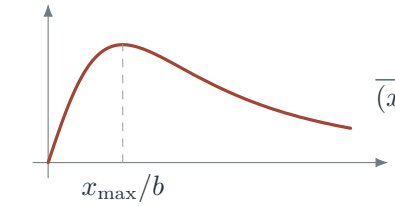


图 2-50　题图　内、外半径限定的带电环面

图示 $a=2b$，$\sigma>0$，纵轴作比例缩放



$$E_x'=0 \Longrightarrow \frac{b^2}{(x^2+b^2)^{3/2}}=\frac{a^2}{(x^2+a^2)^{3/2}}$$

原点与无穷远场为零；唯一正驻点是最大值

图 2-68　解答分析图　求导确定唯一正轴极值

2-T10　一个分量为零的方向

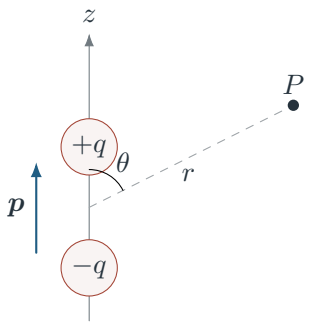
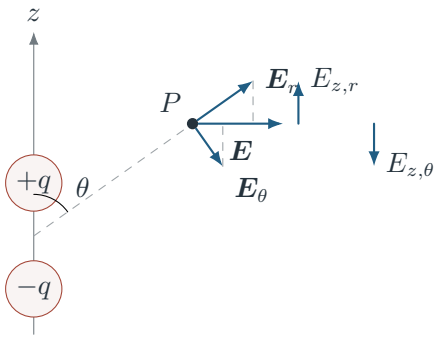


图 2-51　题图　沿 z 的偶极矩与观察角



$\cos^2\theta = 1/3$：两个 z 投影等大反向，总场水平

图示上半锥 $\theta = 54.7356°$，仍有 $\boldsymbol{E} \neq 0$

图 2-69　解答分析图　指定 z 分量为零并不使总场消失

2-T11　矩形通量与立方体面的联系

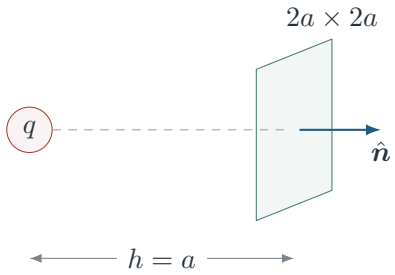
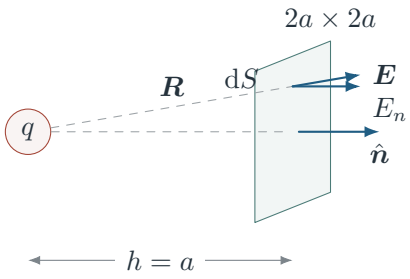
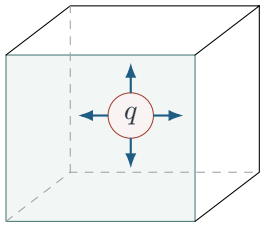


图 2-52　题图　图示方形面；取 $h = a$ 且两半边长为 a

$\mathrm{d}\Phi = E_n \mathrm{d}S = kq\,\mathrm{d}\Omega$





通量按面积分；面上场强一般不均匀

左：方形面立体角；右：以该面补成立方体，点源在中心

图 2-70　解答分析图　方形面是边长 2a 的立方体一面；中心点源向六面等分通量。

2-T12　面外电荷与电场通量

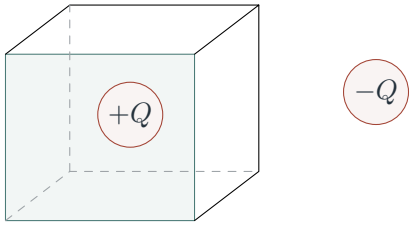
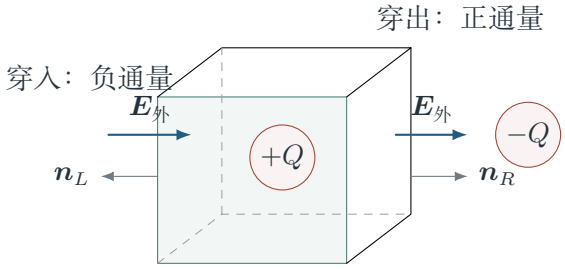


图 2-53　题图　立方体内外的两个电荷



$\Phi_{外}=0$：整个闭面相消，局部场并不为零

图 2-71　解答分析图　局部场含外部电荷贡献，净通量只计内部

2-T13　挑战：有孔立方体的开曲面通量

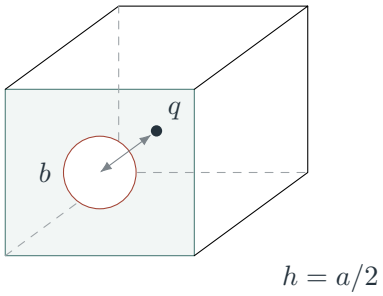
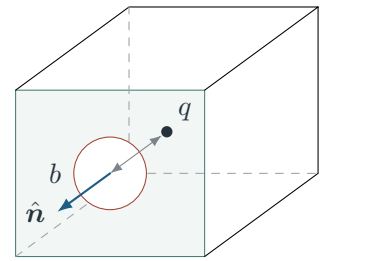


图 2-54　题图　中心电荷与一面中心的圆孔



$$\Phi_{\text{剩余}} = \Phi_{\text{闭合}} - \Phi_{\text{圆盘}}$$

图 2-72　解答分析图　补上辅助圆盘后再从总通量中扣除

2-T14　挑战：体电荷、点电荷和面电荷同时反演

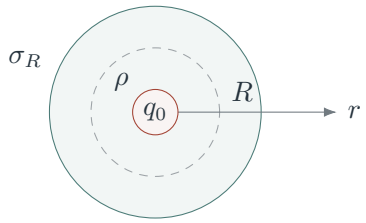


图 2-55　题图　原点、球体及球面三种电荷

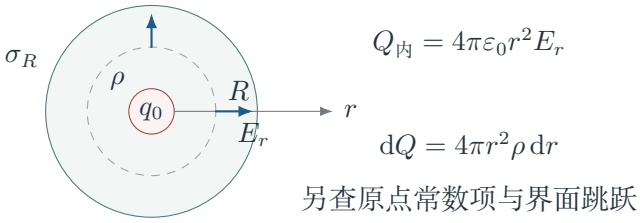


图 2-73　解答分析图　分别读取常数项、导数及界面跳跃

2-T15　挑战：四电荷的方向性抵消

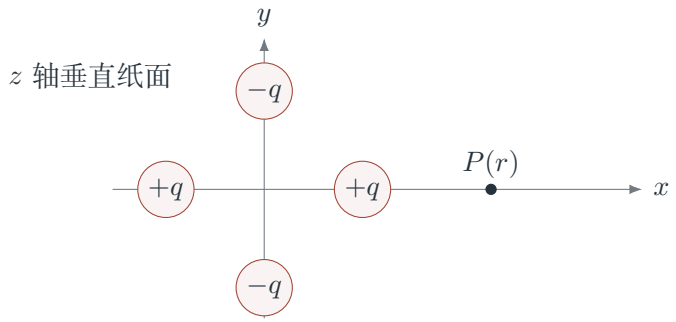
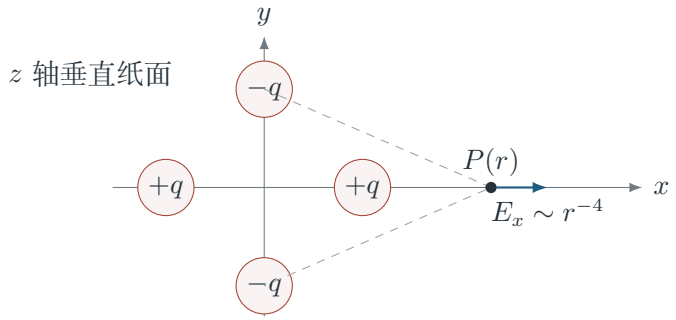


图 2-56　题图　两对异号电荷与特殊观测轴



z 轴全部分量抵消；x 轴保留四极项

图 2-74　解答分析图　不同方向分别处理抵消与展开

2-T16　挑战：线性恢复项被抵消

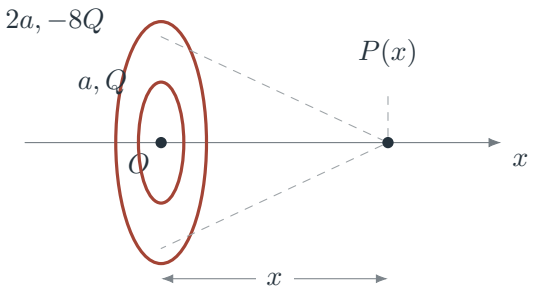
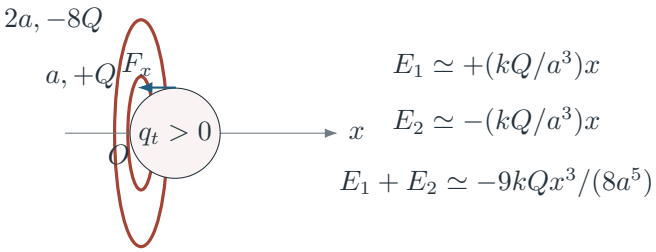


图 2-57　题图　同心共面圆环与轴线粒子



$|x| \ll a$：一次项抵消，三次项使正粒子恢复

图 2-75　解答分析图　线性项抵消后保留三次恢复项

2-T17　挑战：油滴仍在向下运动

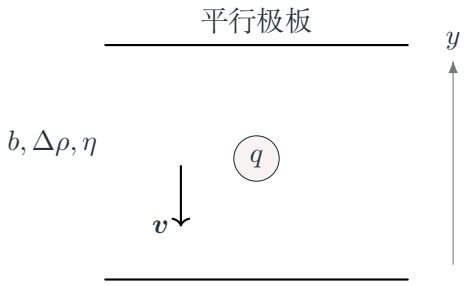
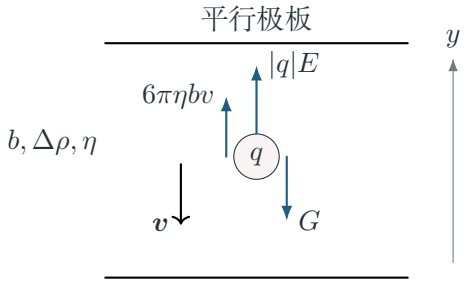


图 2-58　题图　加场后仍向下运动的油滴



仍向下：阻力向上

图 2-76　解答分析图　重力向下，电力与黏滞阻力均向上

2-T18　挑战：量子性数据的参数拟合

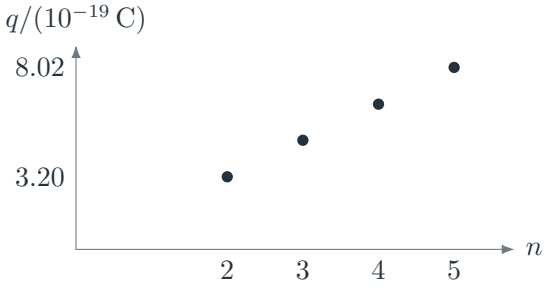
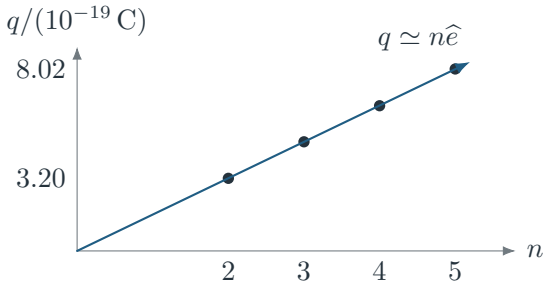


图 2-59　题图　给定整数阶次的四次电荷测量



斜率由最小二乘确定；逐点检查残差

图 2-77　解答分析图　用原始数值拟合斜率，并检查残差